

Response to the Editor and Reviewers

Manuscript title: Multidimensional routine hematologic modeling for acute myocardial infarction phenotype discrimination in coronary artery disease: a single-center retrospective study

Journal: BMC Cardiovascular Disorders

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Dear Editor and Reviewers,

We thank you for the careful assessment. We have made targeted revisions and respond point by point below. The analyzed hematologic measurements were associated with the target hospitalization, although exact within-hospital CBC sampling times could not be reconstructed uniformly. The temporal sensitivity analysis used deidentified hospitalization/coronary-angiography dates from the study period and is not formal temporal validation.

Editor Comments

Editor Comment 1

Comment: Please clarify which complete blood count measurement was selected for each patient: the first measurement on admission, a measurement obtained before coronary angiography or the AMI diagnosis, or another measurement during hospitalization. Where possible, the authors should verify the timing of CBC sampling, hospital admission, coronary angiography, the initial AMI diagnosis, and fibrinogen sampling against the original database. If complete timing information is unavailable, the existing measurement-selection rule should at least be stated explicitly.

Response: During revision, we rechecked the source provenance. The hematologic measurements used in the analysis were obtained in association with the target hospitalization. However, historical timestamp fields in the exported research dataset contained legacy registration inconsistencies and could not reliably reconstruct the exact timing of sampling relative to admission, coronary angiography, or AMI diagnosis for every patient. We therefore do not describe the measurements as uniformly admission-first, pre-angiography, or pre-diagnostic, and we have added the exact record-selection rule to the Methods.

Changes in manuscript: The target-hospitalization association, record-selection rule, and residual timing limitation are stated in Methods and Discussion. (page 6, paragraph 1 of Predictors and laboratory measurements; page 21, paragraph 6 of Discussion).

Editor Comment 2

Comment: The accuracy of the AMI diagnostic labels requires further clarification. The authors should explain whether AMI classification was based solely on discharge diagnoses or also incorporated clinical findings, electrocardiography, cardiac biomarkers, or coronary imaging, as applicable. They should also clarify whether the diagnoses underwent any independent clinical review and acknowledge the limitations of the available diagnostic information.

Response: AMI phenotype was classified from discharge-diagnosis text. Independent clinical adjudication was not available. Cardiac injury markers were supportive only because assay reference limits, serial changes, and measurement timing were incomplete; they were not used to redefine the full cohort. We have clarified these limits and distinguished the strict discharge-text sensitivity definition from clinical adjudication.

Changes in manuscript: The phenotype source, operational definitions, supportive role of biomarkers, and lack of independent adjudication are specified in Methods and reiterated in the Discussion. (page 5, paragraph 1 of Outcome definition).

Editor Comment 3

Comment: The Methods section states that 2548 patient encounters were initially extracted, yielding 2279 unique patients. The authors should account for the 269 records removed during this process and distinguish repeat admissions from exclusions for other reasons. Furthermore, retaining the admission with the most complete clinical data may affect cohort composition. The authors should explain this selection rule and its rationale and revise the Figure 1 flowchart accordingly.

Response: The source export contained 2,548 records representing 2,279 unique patients. Of these, 2,010 had one source record and 269 had two; thus 269 surplus records were removed when retaining one patient-level record per person. The source does not establish that these were distinct admissions. The retained record followed the stated completeness-based priority rule. Figure 1 now distinguishes source records, unique patients, phenotype groups, and the final analysis cohort.

Changes in manuscript: Methods and Figure 1 report the record-to-patient reconciliation and selection rule; the figure does not label the records as repeat admissions. (page 4, paragraph 1 of Eligibility and study population).

Editor Comment 4

Comment: The internal validation procedure requires further technical clarification. The authors should specify how predictions generated for each patient across 10 repetitions of five-fold cross-validation were aggregated; whether identical outer-fold partitions were used across models; and whether hyperparameter tuning, imputation, and standardization were confined to the training folds. They should also clarify whether the 1,000 bootstrap iterations resampled fixed cross-validated predictions or repeated the entire modeling procedure. This distinction affects the sources of uncertainty represented by the reported 95% confidence intervals.

Response: We clarified that stratified five-fold outer cross-validation was repeated 10 times, with five-fold tuning within each outer training set. Each patient's 10 held-out probabilities were averaged for evaluation. Paired models used identical outer partitions where applicable. Imputation, transformations, scaling, and tuning were confined to training data. The 1,000 bootstrap resamples used fixed patient-level mean predictions and outcomes without refitting; accordingly, the intervals do not capture full model-development uncertainty.

Changes in manuscript: The aggregation, shared partitions, training-only processing, and fixed-prediction bootstrap are described in Methods. (page 7, paragraph 1 of Internal validation and uncertainty).

Editor Comment 5

Comment: The temporal sensitivity analysis should be described more precisely. Was the model trained on earlier-period data and evaluated on later-period data, or was cross-validation repeated within the selected temporal sensitivity cohort? These approaches should not be described interchangeably. The authors should report the temporal split point, buffer rules, sample sizes, and numbers of AMI cases. They should also explicitly acknowledge that the difference in AUC between Core-7 and PIV was only 0.002 in this analysis, with a 95% confidence interval that included zero. The modest advantage observed in the primary analysis was therefore not supported to the same extent in the temporal sensitivity analysis.

Response: We rechecked the source provenance and confirm that the temporal sensitivity analysis used hospitalization/coronary-angiography dates from the 2020-2026 study period after patient-specific shifts within +/-182 days; anomalous legacy WBC_test_time fields were not used for temporal allocation. The six-month buffer was excluded. The retained analysis record identifies earlier development, buffer, and later fixed-evaluation groups and preserves their counts: development N=1,001 (AMI=196), buffer N=271 (AMI=79), and later N=548 (AMI=178). However, it does not retain a source-verified calendar cutpoint. The calendar cutoff and date ranges in the prior draft were stale and have been removed; we have not inferred a replacement date. In the later group, Core-7, PIV, and Enhanced AUCs were 0.701, 0.699, and 0.719, respectively; Core-7 minus PIV Δ AUC was 0.002 (95% CI -0.021 to 0.025), which includes zero. The incremental advantage was attenuated. We describe this as a temporal sensitivity analysis, not temporal validation.

Changes in manuscript: The confirmed date source, six-month buffer, frozen group counts and later-cohort estimates are reported in Methods, Results and Figure 4. The calendar cutpoint could not be verified from the retained analysis record, so the stale date has been removed rather than replaced with an inferred value. The temporal sensitivity analysis is not formal temporal validation. (page 16, paragraph 1 of Temporal and diagnostic sensitivity analyses; Figure 4; page 8, paragraph 1 of Temporal sensitivity analysis).

Editor Comment 6

Comment: The primary analysis included 453 patients with AMI, whereas the high-specificity analysis retained only 138 AMI cases. The authors should specify the inclusion and exclusion criteria, explain this substantial reduction in sample size, and clarify whether any eligibility criteria depended on information obtained after the AMI diagnosis. A separate cohort-selection flowchart in the supplementary materials would improve transparency.

Response: The high-specificity comparison retained 138 strict-text AMI cases from the 178 later-group AMI cases and included 288 strict CAD/angina controls (N=426). The other 40 later-group AMI cases did not meet the text rule; they are not described as adjudicated misclassifications. This was a discharge-text sensitivity analysis rather than independent clinical adjudication. A separate flowchart is provided in Additional file 1, Figure S2.

Changes in manuscript: The selection and interpretation are clarified in Results; the supplementary flowchart shows the high-specificity comparison. (page 16, paragraph 1 of Temporal and diagnostic sensitivity analyses; Additional file 1, Figure S2).

Editor Comment 7

Comment: The analysis of missing fibrinogen values should be strengthened. In particular, the authors should report fibrinogen missingness separately for the AMI and non-AMI groups and clarify whether the complete-case comparison between Core-7 and Enhanced was performed using identical patients and validation partitions. If these analyses have already been conducted, the existing results and methods should be reported more clearly.

Response: Fibrinogen was missing in 115/1,820 patients (6.32%), including 27/453 AMI patients (5.96%) and 88/1,367 non-AMI CAD patients (6.44%). Core-7 and Enhanced were compared in the same fibrinogen-complete 1,705 patients and on identical validation partitions; fibrinogen was not imputed for this comparison.

Changes in manuscript: Methods reports group-specific missingness; Results reports the same-sample complete-case comparison on identical validation partitions. (page 9, paragraph 1 of Phenotype and missingness sensitivity analyses).

Editor Comment 8

Comment: To assess incremental performance more completely, I recommend reporting the AUC, Brier score, and paired differences in AUC for Clinical only, Clinical+PIV, Clinical+Core-7, and Clinical+Enhanced, using the same patients and validation partitions. The authors should also specify how missing clinical variables, including age, hypertension, and diabetes, were handled. Reporting only that Clinical+Core-7 achieved a higher AUC than Clinical+PIV does not fully establish the incremental value of the hematologic models over the baseline clinical model.

Response: On the same fibrinogen-complete sample (N=1,705; AMI=426), AUCs were 0.546 for Clinical, 0.717 for Clinical+PIV, 0.730 for Clinical+Core-7, and 0.741 for Clinical+Enhanced. Corresponding Brier scores were 0.187, 0.166, 0.162, and 0.161; the Clinical+Core-7 versus Clinical+PIV paired Δ AUC was 0.0130 (95% CI 0.0004-0.0251). Age and sex were complete; hypertension was missing in 79/1,820 (4.34%) and diabetes in 48/1,820 (2.64%), with both imputed using training-fold medians. Smoking was unavailable. The full paired comparisons are in Additional file 2, Table S8.

Changes in manuscript: The same-sample four-model results, clinical missingness, and paired interval are reported in Methods/Results and Additional file 2, Table S8. (page 15, paragraph 1 of Clinical incremental value and fibrinogen enhancement; Additional file 2, Table S8; page 7, paragraph 1 of Model development and preprocessing).

Editor Comment 9

Comment: The authors should clarify which cross-validated predictions were used to calculate the calibration intercepts and slopes reported in Figure 3 and Table 2. Displaying the distribution of predicted probabilities or the number of patients within each calibration group would also aid interpretation. In addition, the authors should explain why some sensitivity analyses in Figure 5 are presented with AUC point estimates but no 95% confidence intervals.

Response: Calibration intercepts, slopes, and plots were calculated from the patient-level arithmetic mean of the 10 held-out predictions. Additional file 2, Table S6 reports the calibration-group sizes, AMI counts, and predicted-probability summaries. The available analysis output provided some alternative date-shift estimates only as point estimates; confidence intervals were unavailable, so we report them as such rather than implying interval estimation.

Changes in manuscript: The prediction source and group-level counts are identified in Results and Additional file 2, Table S6; Figure 5 and its legend identify estimates without intervals. (page 12, paragraph 1 of Internal model performance; Additional file 2, Table S6).

Editor Comment 10

Comment: Finally, the conclusions should reflect the magnitude and robustness of the observed improvements. In the primary analysis, the increase in AUC= 0.017 for Core-7 versus PIV and 0.010 for Enhanced versus Core-7. Both improvements were modest, and the incremental advantage of Core-7 was further attenuated in the temporal sensitivity analysis. The authors should therefore emphasize that jointly modeling routine hematologic measurements provided additional information for AMI phenotype discrimination, rather than suggesting that the model is ready for clinical diagnosis or triage. The existing qualifications in the Discussion should be retained, and the conclusions should be revised consistently after the methodological issues have been addressed.

Response: We revised the abstract, Discussion, and Conclusions to characterize the gains as modest additional information for AMI phenotype discrimination. The Core-7 increment over PIV was attenuated in the temporal sensitivity analysis, and the findings do not establish diagnostic or triage readiness. The existing qualifications are retained and strengthened where needed.

Changes in manuscript: The revised claim appears in the Abstract, Discussion, and Conclusions. (page 21, paragraph 1 of Conclusions).

Additional editorial ethics requirement

Comment: Authors reporting studies involving humans/human data/the use of human samples must confirm that all experiments were performed in accordance with relevant guidelines and regulations. Please ensure that a statement to confirm this has been included in the 'Ethics approval and consent to participate' section.

Response: We added the requested statement that all methods were performed in accordance with relevant guidelines and regulations. The Ethics section identifies approval no. K202602-10, and the consent statement now records the ethics-committee waiver for this retrospective study.

Changes in manuscript: The Ethics approval and Consent to participate declarations are corrected; Consent for publication is stated as not applicable. (page 22, paragraph 1 of Ethics approval).

Reviewer 1

Major Comment 1

Comment: Validation strategy and generalizability

The manuscript describes repeated stratified five-fold cross-validation, but it is unclear whether a formal nested cross-validation procedure was used. Please clarify how the outer validation loop and inner hyperparameter-tuning loop were performed, and whether all preprocessing, imputation, and parameter selection were conducted within the appropriate training folds. Since no independent external cohort appears to have been used, the generalizability of the model remains uncertain. The authors should clarify whether the temporal analysis represents a true temporal validation and moderate claims regarding external applicability.

Response: We specified the nested five-fold outer/inner procedure, training-fold-only preprocessing and tuning, and patient-level averaging of the 10 held-out predictions. No independent external cohort was used. The analysis based on deidentified hospitalization/coronary-angiography dates is a temporal sensitivity analysis, not formal temporal validation; generalizability therefore remains uncertain.

Changes in manuscript: Methods and Discussion describe the validation procedure and limitations; Results reports the fixed later-group comparison. (page 7, paragraph 1 of Internal validation and uncertainty).

Major Comment 2

Comment: Potential collinearity among CBC variables

CBC-derived variables may be correlated with each other and with the composite inflammatory indices. Although Elastic-net regression can partly address collinearity, it may still affect variable-selection stability and model interpretation. Please provide a brief assessment of collinearity and discuss the stability of the selected Core-7 variables.

Response: We report the exploratory correlation assessment and selection stability in Additional file 1, Figure S3 and Additional file 2, Table S7. The maximum absolute pairwise Spearman correlation among the Core-7 predictors was 0.469 (neutrophil and monocyte counts). Neutrophils, lymphocytes, monocytes, platelets, and RDW were selected in all resampling iterations; MPV was selected in 99% and hemoglobin in 95%. Elastic-net estimates are interpreted as predictive associations, not isolated biological effects.

Changes in manuscript: The stability summary is reported in Results and Additional file 2, Table S7; predictor correlations are shown in Additional file 1, Figure S3. (page 18, paragraph 1 of Predictor stability; Additional file 1, Figure S3; Additional file 2, Table S7).

Major Comment 3

Comment: Missing data in clinical variables

The manuscript describes strict handling of missing CBC variables, but it is unclear whether the clinical covariates included in the models also had missing values. Please report the extent of missingness and explain how missing clinical variables were handled. The authors should also clarify whether imputation, if used, was performed within the cross-validation procedure.

Response: Age and sex were complete. Hypertension was missing in 79/1,820 patients (4.34%) and diabetes in 48/1,820 (2.64%). Missing clinical values were imputed using medians estimated from each training fold; smoking was unavailable and was not included. Imputation remained inside the validation pipeline.

Changes in manuscript: Methods and Results specify the clinical-variable completeness and fold-specific imputation. (page 7, paragraph 1 of Model development and preprocessing; page 15, paragraph 1 of Clinical incremental value and fibrinogen enhancement).

Major Comment 4

Comment: Incomplete reporting of composite inflammatory indices

The manuscript states that PIV had the best performance among the conventional composite inflammatory indices, but detailed results for the other indices are not provided. Please report the performance of all prespecified indices, including their AUCs and confidence intervals, and clarify whether Core-7 was compared with each index or only with PIV.

Response: We report AUCs and 95% confidence intervals for all prespecified conventional indices in Additional file 2, Table S4. Each index was evaluated as a continuous predictor in the repeated nested validation framework. PIV was the prespecified primary comparator for the paired Core-7 comparison; the other indices are descriptive benchmarks, and no additional pairwise Core-7-versus-index tests were performed.

Changes in manuscript: The Methods now distinguishes the prespecified PIV comparison from descriptive benchmark indices; all benchmark AUCs and confidence intervals are in Additional file 2, Table S4. (page 6, paragraph 1 of Conventional inflammatory indices; Additional file 2, Table S4).

Major Comment 5

Comment: Selection of records for patients with repeated admissions

For patients with repeated hospitalizations, the manuscript states that the most complete record was selected. This approach may introduce selection bias. Please describe how “most complete” was defined and report the number of patients with repeated admissions. The prevalence of prior coronary artery disease, previous myocardial infarction, and prior revascularization should also be reported and their potential role as predictors should be clarified.

Response: The source export contained 2,279 unique patients; 269 had two source records, but the data do not establish that these were distinct admissions. One record per patient was selected using the stated completeness-based priority rule. Validated index-relative fields for prior CAD, MI, PCI, or CABG were unavailable, and diagnosis mentions could not establish prior-history prevalence. We therefore did not report unsupported prevalences or include these as predictors; this limitation is now explicit.

Changes in manuscript: The patient-level flow and selection rule are shown in Methods/Figure 1, and unavailable prior-history fields are acknowledged in the Discussion. (page 4, paragraph 1 of Eligibility and study population).

Major Comment 6

Comment: Fairness of the comparison with PIV

Core-7 was developed as a multivariable model, whereas PIV appears to have been analyzed as a single continuous variable. Please clarify how PIV was modeled, including whether transformation, standardization, or calibration was applied. All comparator indices should ideally be evaluated using the same validation framework to ensure a fair comparison.

Response: PIV was modeled as a single continuous predictor after $\log(1+PIV)$ transformation and standardization using training-fold parameters, in an unpenalized logistic model. It was evaluated with the same patients and outer-fold partitions as Core-7 where applicable. No post hoc calibration was applied; all reported performance derives from held-out predictions.

Changes in manuscript: The PIV transformation and shared validation framework are specified in Methods; comparative estimates are reported in Tables 2 and S4. (page 6, paragraph 1 of Conventional inflammatory indices).

Major Comment 7

Comment: Clinical relevance and comparison with standard biomarkers

The manuscript should better clarify the clinical role of Core-7. High-sensitivity cardiac troponin is the standard biomarker for AMI diagnosis and is generally available within a short time. If troponin data are available, the authors should evaluate whether Core-7 provides incremental value beyond troponin. If troponin was not available, this limitation should be clearly acknowledged, and the authors should explain the specific clinical scenario in which Core-7 would be useful.

Response: A standardized incremental comparison beyond high-sensitivity troponin was not possible because assay platforms, reference limits, serial change, and timing were not sufficiently harmonized. Troponin and CK-MB served only as supportive phenotype checks, not as a complete adjudication standard. We now state that the model was not evaluated for real-time triage, treatment decisions, or performance before biomarker results; any future role would require independent, time-anchored evaluation.

Changes in manuscript: The troponin-comparator limitation and absence of diagnostic/triage readiness are stated in the Discussion. (page 20, paragraph 5 of Discussion).

Minor Comment 1

Comment: Patient selection

The number of included patients over the six-year study period appears relatively low for a tertiary hospital. Please clarify whether patients were screened consecutively. The possibility of selection bias related to data completeness should be discussed.

Response: A complete eligible-patient screening log was unavailable, so consecutive enrollment could not be confirmed. We state this explicitly and discuss possible selection related to data completeness and diagnosis

availability. The source population and target-hospitalization association of the analyzed hematologic measurements have been clarified; exact CBC sampling times remain incompletely reconstructable.

Changes in manuscript: The study-design paragraph and limitations now state that consecutive screening cannot be confirmed and identify selection concerns. (page 4, paragraph 1 of Study design and setting).

Minor Comment 2

Comment: Timing of CBC measurement

The manuscript should specify when the CBC was collected in relation to hospital admission, symptom onset, troponin testing, coronary angiography, and treatment. If multiple CBC measurements were available, please describe the criteria used to select the index measurement.

Response: The hematologic measurements used in the analysis were obtained in association with the target hospitalization. Historical timestamp fields contained legacy registration inconsistencies and could not reliably establish the exact timing relative to admission, angiography, or AMI diagnosis for every patient. The stated selection rule is completeness-based, not a clinical-time hierarchy; we therefore make no claim that measurements were uniformly admission-first, pre-angiography, or pre-diagnostic. Symptom-onset timing was not available.

Changes in manuscript: The CBC measurement paragraph states the selection rule and timing limitations. (page 6, paragraph 1 of Predictors and laboratory measurements).

Minor Comment 3

Comment: Informed consent

The manuscript states that informed consent was obtained from all participants. Please clarify how consent was obtained for the entire study population, particularly given the retrospective design and the fact that the study was conducted in 2026, when some patients may have died or may no longer have been able to provide consent. The authors should specify whether consent was obtained from legally authorized representatives or whether another ethics-approved procedure was used, and ensure that the description is consistent with the ethics approval.

Response: The ethics statement now specifies that informed consent was waived by the Ethics Committee because of the retrospective study design; the prior statement that all participants provided consent has been removed.

Changes in manuscript: Declarations, Ethics approval and Consent to participate. (page 22, paragraph 1 of Consent to participate).

Reviewer 2

Comment 1

Comment: The authors have conducted a comprehensive study to develop multidimensional model to discriminate AMI phenotypes. The research has high methodological quality and provides clinically relevant information. I would suggest to add decision-curve analysis to strengthen the results.

Response: We added exploratory decision-curve analysis for the four clinical models using the same fibrinogen-complete sample (N=1,705) and held-out predictions, with treat-all and treat-none references across thresholds of 0.05-0.50. The analysis is supplementary and is not presented as evidence of clinical utility or treatment benefit.

Changes in manuscript: Methods and Results describe the analysis; the curve is in Additional file 1, Supplementary Figure S1. (page 9, paragraph 1 of Reporting and reproducibility; Additional file 1, Supplementary Figure S1).

Sincerely,
The Authors